

Hydrogen Fuel Assist for Logistics & Fleet Vehicles

5–18% fuel savings across your fleet. Up to 95% PM reduction. No engine modification. Just add water.

THE SCALE OF THE PROBLEM

India's logistics sector is worth \$250 billion and contributes 13–14% of GDP — **but it is also among the most carbon-intensive logistics ecosystems in the world.** Road transport handles 70% of all freight movement, with diesel trucks, construction & mining vehicles contributing almost 50% of total transport CO₂ emissions. Fuel typically accounts for close to 50% of a fleet operator's total operating costs.

Over 90% of medium and heavy commercial vehicles (MHCVs) run on diesel. Over 85% of India's truck fleet is owned by individual operators or small fleet owners with limited access to expensive EV or hydrogen transitions. Electric trucks cost 20–50% more than diesel equivalents, charging infrastructure barely exists for heavy-duty routes, and green hydrogen is years away from cost parity.

The industry needs a solution that works today, on existing vehicles, with immediate financial returns — not one that requires replacing the entire fleet.

THE SGT GREENDRIVE SOLUTION

SGT HydroEdge has developed GreenDrive — a proprietary, patent-applied HHO (oxyhydrogen) generation system that retrofits to any diesel vehicle engine. GreenDrive generates hydrogen-oxygen gas on-demand through water electrolysis and injects it into the engine's air intake. Hydrogen's flame propagation speed — roughly 10x faster than diesel vapour — and extremely low ignition energy act as a combustion catalyst: the diesel burns more completely, less fuel is wasted, and emissions are dramatically reduced at source.

No engine modification. No exhaust system changes. No downtime for installation. No backpressure. No N₂O creation. No urea or chemicals. **Input: water and electricity from the vehicle's own alternator. Output: significant fuel savings, cleaner air, and extended engine life through continuous decarbonisation.**

SGT'S OWN TESTING & VALIDATION

SGT HydroEdge has conducted extensive testing on vehicle engines over three years of development.

- **KOEL (Kirloskar Oil Engines)** partnership — active testing and validation on diesel engines
- **ARAI (Automotive Research Association of India)** — conducting joint testing for Government of India approval as emission reduction technology
- **On-road fleet trials** conducted across trucks and commercial vehicles demonstrating consistent fuel savings and emission reductions in real-world operating conditions and learned & improved with failures
- **Engine decarbonisation validated** — SGT's research confirms that continuous HHO-assisted combustion cleans internal carbon deposits during normal operation, extending engine life and reducing maintenance

PROVEN RESULTS

5–18%

Fuel Savings

Up to 95%

PM 2.5 Reduction

>50%

CO Reduction

>46%

HC Reduction

7–20%CO₂ Reduction

WHY GREENDRIVE — NOT POST-TREATMENT RETROFITS

Most fleets today run vehicles with OEM-fitted after-treatment systems like DPF, SCR, and DOC. While these systems reduce regulated pollutants, they add backpressure, regeneration cycles, and dosing loads — increasing fuel consumption and operating costs. GreenDrive is fundamentally different. It works upstream by improving combustion efficiency inside the engine. Better combustion reduces fuel use and CO₂ at the source — making existing OEM systems perform more efficiently without adding complexity or load.

	Post-Treatment Retrofits	SGT GreenFleet
Backpressure	Creates exhaust restriction — engine works harder, uses MORE fuel	Zero backpressure — nothing touches exhaust
Fuel impact	Increases consumption 2–5% due to backpressure	REDUCES consumption 3–15%
N2O creation	SCR generates N2O — a GHG 265x worse than CO2	Zero N2O — no catalytic chemicals
CO2 reduction	No CO2 reduction. More fuel = more CO2	Less fuel = directly less CO2
Maintenance	DPF cleaning, urea/AdBlue refills, catalyst replacement	Water only. Simple. Inexpensive.
Engine modification	Exhaust system modification. Warranty risk.	Zero modification. Air intake retrofit only.
Engine life	No engine benefit. Backpressure increases wear.	Continuous decarbonisation extends engine life.
Approach	Treats symptoms AFTER combustion	Fixes root cause: incomplete combustion

THE FINANCIAL IMPACT FOR FLEET OPERATORS

Fuel is 45–55% of total fleet operating cost. Even a 5% fuel saving across a fleet creates massive financial impact:

- **A fleet of 100 trucks** averaging 4 km/litre, running 200,000 km/year each, consumes ₹42.5 crore of diesel annually (at ₹85/litre). A 5% saving = ₹2.12 crore/year. A 10% saving = ₹4.25 crore/year.
- **A single long-haul truck** spending ₹30–40 lakh on diesel per year saves ₹1.5–4 lakh/year with GreenFleet — payback in months, not years.
- **Mining vehicles and construction equipment** burning 80–200 litres/day save ₹2,000–8,500 per vehicle per day at 10% fuel reduction.
- **Extended engine life and reduced maintenance** from continuous decarbonisation add additional savings that compound over time.

PATENT-APPLIED TECHNOLOGY

High HHO Yield	Proprietary cell design delivers significantly higher gas output per unit of electrical input than conventional HHO generators
Low Power Draw	Optimised to minimise load on vehicle alternator, ensuring net energy benefit even in city driving conditions
Vibration & Heat Resistant	Engineered for the harsh conditions of commercial vehicles — road vibration, engine heat, dust, and extreme temperatures
Safety by Design	On-demand generation. Zero hydrogen storage. Flashback arrestors, pressure relief, fail-safe shut-off on engine stop
Easy & Green Maintenance	Driver or workshop serviceable. No specialist tools. Input: water and standard electrolyte only. Monitored remotely for system health with inbuilt IoT to reduce cost of maintenance & additional CO2 footprint for servicing.
Compact & Vehicle-Ready	Small footprint designed to mount under chassis of commercial vehicles without structural modification

DIGITALLY CONNECTED — IOT & FLEET MANAGEMENT

Every GreenDrive unit is fully digitally connected.

Built-in IoT sensors monitor system performance, HHO output, and vehicle operating parameters in real time. Data feeds into GreenVision — SGT's AI-powered Digital Twin platform.

- **Per-vehicle fuel tracking** — actual savings quantified per truck, per route and per Driver
- **Fleet-wide dashboard** — monitor every equipped vehicle across the fleet from one interface
- **Emission reporting** — CO₂, PM, CO, HC reductions documented for BRSR compliance, ESG disclosures, and carbon credit documentation
- **Driver behaviour insights** — correlate driving patterns with fuel efficiency to identify best practices
- **Predictive maintenance** — detect HHO system and engine degradation before failures, reducing unplanned downtime and roadside breakdowns

WHAT GLOBAL RESEARCH PROVES

HHO-enriched combustion in diesel engines — the same engines powering commercial vehicles — has been extensively validated by researchers worldwide:

FUEL EFFICIENCY & PERFORMANCE

- Diesel fuel consumption reduced by 11.7–12.3% in standardised road testing (NEDC cycle) on commercial vehicles, with THC, CO₂, CO, and smoke opacity emissions all reduced simultaneously
- BSFC (brake specific fuel consumption) reduced by 12.6–13.7% on average across controlled engine testing, with some studies documenting up to 20% reduction at optimised HHO flow rates
- Real-world vehicle road tests (40 km extra-urban driving cycles) confirm laboratory results translate directly to operational fuel savings under actual driving conditions
- Engine thermal efficiency improvements of up to 31.5%, meaning more power extracted per litre of diesel — engines deliver same output with less fuel, or more output with same fuel
- 500-hour continuous dual-combustion study confirmed no structural damage to engine components, validating long-term safety and reliability for fleet operations

EMISSION REDUCTIONS

- PM and smoke opacity reduced by 25–95% depending on HHO flow rate and engine load. Smoke opacity reduction of 26.8% documented under NEDC driving cycle
- CO emissions reduced by 13–54.5% across multiple studies. HC emissions reduced by 7.7–46.15%. CO₂ reduced by 7–15.2%
- NO_x reductions of 4.3–15.5% documented at moderate loads. SGT's optimised calibration targets consistent NO_x reduction across operating conditions
- **Zero N₂O creation** — unlike SCR post-treatment systems that generate nitrous oxide (265x more potent than CO₂ as a greenhouse gas), SGT GreenDrive produces no secondary greenhouse gases
- Engine decarbonisation reduces soot formation at source, meaning cleaner exhaust from day one with progressively cleaner internals over time

Research findings based on work by scientists at institutions worldwide including University of Technology Baghdad, UNAM Mexico, Escuela Superior Politécnica del Litoral Ecuador, multiple research groups in India, Turkey, Egypt, and Malaysia, published in peer-reviewed journals indexed in ScienceDirect, Springer, MDPI, SAGE, Elsevier, and Nature Scientific Reports. SGT's own testing adds proprietary validation data. Full references available on request.

APPLICATIONS ACROSS THE LOGISTICS, MINING & CONSTRUCTION ECOSYSTEM

Long-Haul Trucks

Inter-state freight, heavy commercial vehicles

Construction

Excavators, loaders, site transport

City Distribution

Last-mile delivery, LCVs, e-commerce fleets

Bus Fleets

State transport, private operators, staff buses

Mining Vehicles

Dumpers, tippers, haul trucks, site equipment

Agriculture & Tractors

Farm equipment, diesel pump sets



Vehicle types	Trucks, buses, tippers, dumpers, excavators, tractors, LCVs — any diesel vehicle running on 12V or 24V power
Installation	Retrofit to air intake and DC power from vehicle battery pack.. No engine modification. Minimal downtime.
Hydrogen storage	None — on-demand generation. Zero storage risk. No ADR/ PESO compliance needed.
Input required	Water and electricity from vehicle alternator. No chemicals, no urea.
Validation	ARAI Govt. of India testing + on-road fleet trials. Working with a leading OEM for validation on new engines.
Connectivity	IoT enabled. SGT GreenVision fleet management Digital Twin platform.
Technology	Patent-applied. Vibration-resistant. High yield. Low power.

Ready to Cut Your Fleet's Fuel Bill & Carbon Footprint?

Ask us for a no-obligation fleet assessment. We'll show you the numbers.

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